

Series Temporales

Matrix Profile

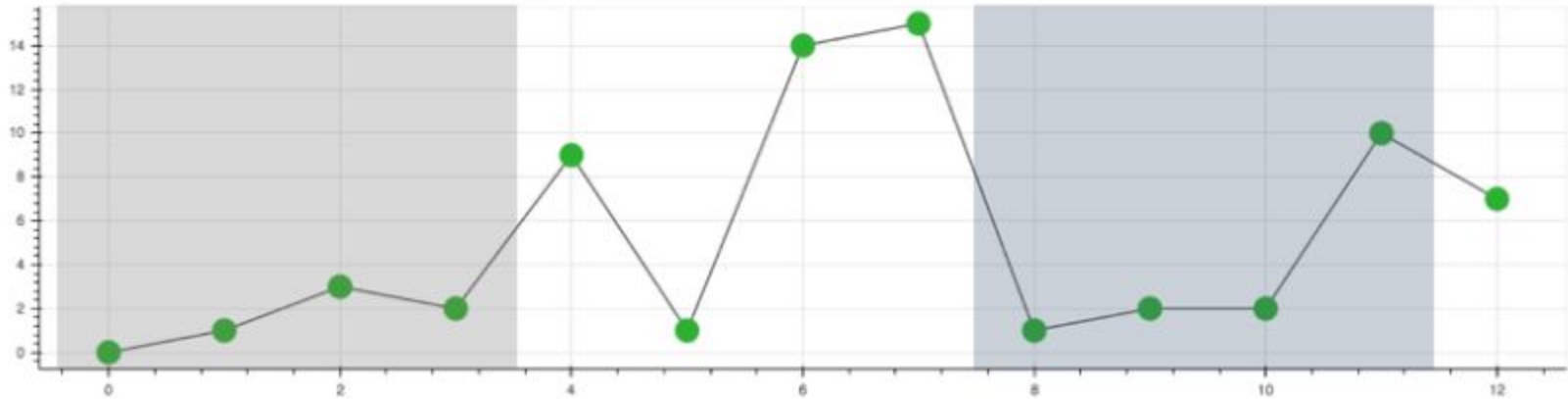


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Comparación de sub-secuencias



0	1	3	2	9	1	14	15	1	2	2	10	7
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Euclidean
Distance

$(0,1,3,2)$

$(1,2,2,10)$

$$D = \sqrt{(1-0)^2 + (2-1)^2 + (2-3)^2 + (2-10)^2}$$

$$D = \sqrt{67}$$

Pairwise Euclidean Distance

0	1	3	2	9	1	14	15	1	2	2	10	7
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#DistanceProfile

Pairwise Euclidean Distance

0	1	3	2	9	1	14	15	1	2	2	10	7
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0	1	3	2	9	1	14	15	1	2	2	10	7
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	7.4	6.9	14.7	19.3	17.7	19.9	15.0	8.2	8.9			
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Distance Matrix

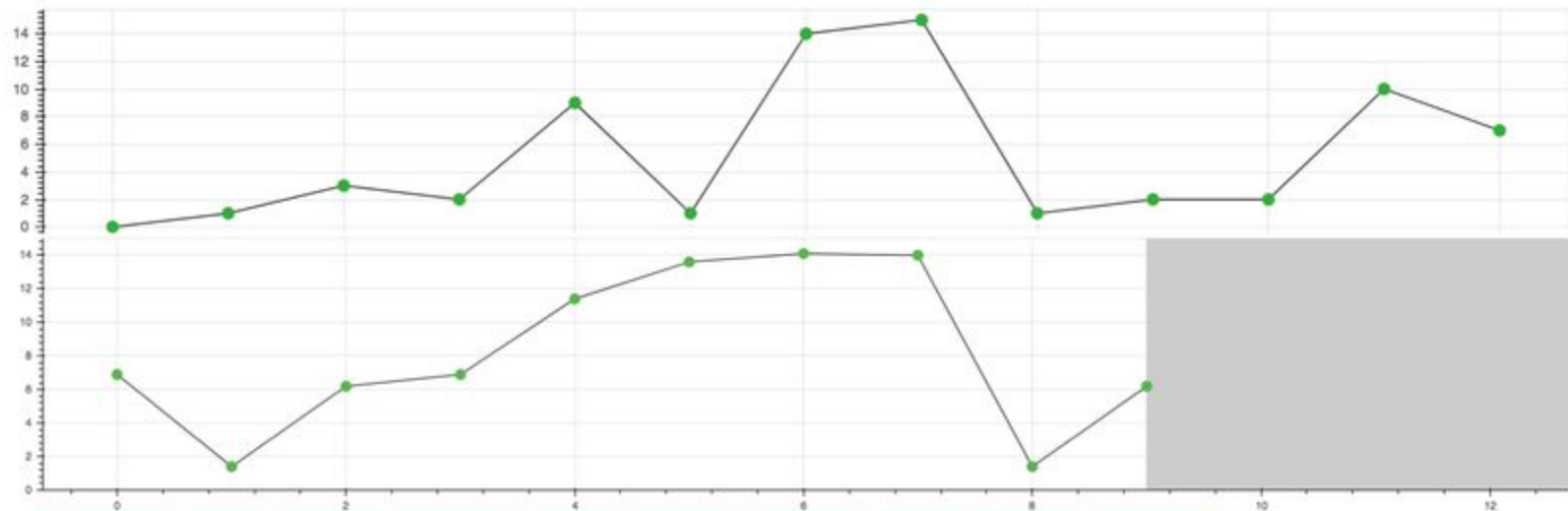
	*	6.9	*	*	*	*	*	*	*			
*		*	*	*	*	*	*	1.4	*			
*	*		*	*	*	*	*	*	6.2			
*	7.9	*		*	*	*	*	*	*			
*	*	*	*		*	*	*	*	11.4			
*	*	13.6	*	*		*	*	*	*			
*	*	*	*	*	*		14.1	*	*			
*	*	14.0	*	*	*	*		*	*			
*	1.4	*	*	*	*	*	*		*			
*	*	6.2	*	*	*	*	*	*				

Matrix Profile

	*	6.9	*	*	*	*	*	*	*			
*		*	*	*	*	*	*	1.4	*			
*	*		*	*	*	*	*	*	6.2			
*	7.9	*		*	*	*	*	*	*			
*	*	*	*		*	*	*	*	11.4			
*	*	13.6	*	*		*	*	*	*			
*	*	*	*	*	*		14.1	*	*			
*	*	14.0	*	*	*	*		*	*			
*	1.4	*	*	*	*	*	*		*			
*	*	6.2	*	*	*	*	*	*				

#DistanceProfiles

0	1	3	2	9	1	14	15	1	2	2	10	7
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6.9	1.4	6.2	7.9	11.4	13.6	14.1	14.0	1.4	6.2
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Demo



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Demo - <https://stumpy-dev.github.io/stumpy-demo/>



Usos



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Usos y ejemplos

Muchos ejemplos y formas de usar en:

The UCR Matrix Profile Page

<https://www.cs.ucr.edu/~eamonn/MatrixProfile.html>

Referencias



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Incluye múltiples papers, slides y links a tutoriales

Se describen distintos posibles usos

Motif Mining in Time Series Data

<https://www.cs.ucr.edu/%7Eeamonn/TimeSeriesMotifs/>

Tutorial y slides más recientes

Stumpy - Biblioteca de python que implementa Matrix Profile

<https://stumpy.readthedocs.io/en/latest/index.html>